

RUTHVIK VISHWANATHWAR

Senior Data Engineer | Azure Data Engineer | Databricks Engineer

ruthvikvishwa@gmail.com • 919-397-3265 • ruthvikvishwanathwar.com

SUMMARY

Senior Data Engineer and **Databricks Certified Data Engineer Associate** with 11 years of experience designing enterprise-scale lakehouse architectures that enable analytics, compliance, and AI-driven decision-making across healthcare and financial services. Expertise in multi-source data orchestration, governance frameworks, and establishing data quality as organizational discipline. Proven track record architecting systems that reduce time-to-insight by 60%+ and position heterogeneous enterprises for ML-driven operations while maintaining HIPAA/PHI controls and audit accountability.

CERTIFICATIONS

Databricks: Databricks Certified Data Engineer Associate, Platform Architect, Certified Platform Administrator, Generative AI Fundamentals, AI Agent Fundamentals, Lakehouse Fundamentals | SQL for Data Science

CORE COMPETENCIES

- **Cloud & Platforms:** Azure Data Factory, Databricks, Synapse Analytics, ADLS Gen2, Blob Storage, Logic Apps, DevOps, Microsoft Fabric, OneLake, Snowflake, Power BI
- **Data Engineering:** ETL/ELT Pipelines, Delta Lake, Medallion Architecture, Data Lakehouse, Data Warehousing, SSIS Modernization, Real-Time Streaming (Spark Structured Streaming), Cosmos DB, Lakehouse Federation, Airflow, DAG design, SLA monitoring, backfills, dependency management.
- **Programming:** Python, PySpark, SQL, T-SQL, DAX, Spark SQL
- **Performance & Optimization:** Z-Ordering, Compaction, Caching, OPTIMIZE/MERGE, Data Skipping, Index Tuning, Query Tuning, Cost Optimization
- **AI & ML:** Generative AI, LLM Data Pipelines, MLflow, Feature Engineering, AI Agent Orchestration
- **DevOps & Tooling:** Git, GitHub, Azure DevOps CI/CD, GitHub Actions, Databricks Notebooks, JIRA, Confluence, ServiceNow
- **Governance & Compliance:** HIPAA/PHI Masking, Access Controls, Schema Enforcement, Audit Logging, SCDs, Data Quality Validation

PROFESSIONAL EXPERIENCE

Senior Data Engineer | *Dotcom-Team, Frisco, TX (Clients: Sentara Health, Georgia Pacific)* May 2022 – Present

- Led migration of 40+ legacy SSIS/SQL Server ETL workflows into modular Databricks PySpark/Spark SQL jobs, retiring legacy execution dependencies and reducing average pipeline runtime by ~60% across production workloads.
- Built sophisticated reconciliation rules framework supporting field mappings, crosswalk logic, effective date/business rules, normalization/tolerance thresholds, and expected difference calculations, enabling semantic alignment across heterogeneous data models while parameterizing tolerance rules for flexible exception detection and reducing false positives by 60%+ through data-driven threshold tuning.
- Designed and enforced Medallion Architecture (Bronze/Silver/Gold) on Delta Lake, implementing ACID transactions, schema evolution, time travel, and versioning to ensure data reliability and auditability across the entire lakehouse.
- Engineered HIPAA/PHI-compliant data pipelines for large-scale healthcare datasets (claims, member, provider), applying column-level masking, role-based access controls, and PHI tokenization in alignment with enterprise governance standards.
- Drove Delta Lake performance optimization across enterprise datasets using Z-ordering, adaptive partitioning, OPTIMIZE/MERGE, and file compaction, improving query response times by up to 70% on multi-billion row tables.
- Reduced a critical batch process from 14 hours to under 3 hours through index tuning, stored procedure refactoring, and query optimization using SQL Server Database Engine Tuning Advisor.

- Established reconciliation landing DB as foundational data asset for downstream ML/AI pipelines by embedding data quality assertions, historical exception patterns, and transformation lineage into Delta Lake, enabling future ML models to learn from past discrepancies, automate anomaly detection, and predict data quality issues before they surface, positioning reconciliation as proactive ML-powered quality control rather than reactive issue response.
- Built near-real-time Spark Structured Streaming pipelines for event-driven ingestion into Delta Lake, supporting checkpointing, schema evolution, late-arriving data handling, and replayable processing semantics.
- Standardized Databricks cluster policies and workload-specific compute configurations while optimizing shuffle volume, partition strategy, and redundant transformations to improve cost control, security, and pipeline efficiency.
- Led technical design reviews and mentored data engineers on Spark optimization, Delta Lake modeling, Unity Catalog governance, and CI/CD standards, improving consistency across lakehouse development and production support practices.

Stack: Azure Databricks, PySpark, Spark SQL, ADF, Delta Lake, Delta Live Tables, SQL Server 2019, Azure Synapse, Power BI, SSIS, Python

Senior Data Engineer | *E-IT Professionals, Livonia, MI (Blue Cross Blue Shield of Kansas)* May 2021 – May 2022

- Implemented data validation and reconciliation checks across source-to-report pipelines, improving trust in eligibility, claims, and provider metrics by identifying missing records, duplicate claims, and inconsistent member attributes before executive reporting.
- Designed orchestration patterns across ADF, Databricks Jobs, and Airflow DAGs, standardizing retries, dependency handling, SLA monitoring, and backfill operations.
- Optimized Databricks workload efficiency through partition strategy, file compaction, job tuning, and cluster configuration standards, reducing compute waste and improving pipeline scalability across high-volume datasets.
- Tuned high-volume SQL workloads through indexing strategy, execution-plan analysis, and stored procedure refactoring, reducing report refresh time by 40% and improving dashboard responsiveness for finance, claims, and operational reporting users.

Stack: SQL Server 2016/2017, Power BI, Power Query, DAX, GitHub Actions, SSMS, JIRA, Confluence, Databricks, Airflow DAGs

Database Developer | *SkillStorm LLC, Addison, TX (Client: Bank of America)* Nov 2019 – Apr 2021

- Built and deployed SSIS packages for multi-source data extraction and transformation, ingesting from Excel, flat files, and relational databases with Conditional Split, Fuzzy Lookup, and Multicast for data scrubbing and staging validation.
- Designed automated ETL processes ensuring end-to-end data consistency, quality, and scalability across multiple operational systems.
- Investigated and resolved performance bottlenecks in stored procedures and complex SQL queries, improving execution plans and reducing load times across key reporting pipelines.

Stack: SQL Server 2016/2017, SSIS, SSRS, SSAS, Power BI, Visual Studio 2017, JIRA, Confluence, Bitbucket

Database Developer | *E-IT Professionals Corp, Livonia, MI (Client: Visa Inc, Vintech)* Aug 2016 – Oct 2019

- Maintained referential integrity and data consistency using SQL constraints; reviewed and refactored complex multi-join queries to improve performance across high-transaction financial datasets.
- Designed automated ETL processes ensuring end-to-end data consistency, quality, and scalability across multiple operational systems.

Stack: SQL Server 2016, SSIS, SSRS, Visual Studio 2017, JIRA, Confluence, Bitbucket

EDUCATION

M.S. Information Technology | GPA: 3.8/4.00 | Campbellsville University, Kentucky

M.S. Computer Science | GPA: 3.7/4.00 | New York Institute of Technology, Manhattan, NY

B.S. Information Technology | GPA: 3.5/4.00 | Scient Institute of Technology, India